

Marine Mattress FAQ

The costs identified in the table below are intended as a guide budget purposes on typical marine mattress projects. Experience has shown that local site conditions, stone cost and labor conditions will impact the price. PRICES TYPICAL IN CONTINENTAL US.

Mattress Thickness	Prefabricated, unfilled cost (delivered)	Prefabricated, filled & staged on site	Filled Mattress installed from land (typical range)	Filled Mattress installed from barge (range)
6-inch	\$4 - \$5 / sq ft*	\$7 - \$10/ sq ft*	\$15 - \$18 / sq ft**	\$18 - \$20 / sq ft**
12-inch	\$6 - \$8 / sq ft*	\$11 - 14 / sq ft*	\$17 - \$20 / sq ft**	\$22 - \$30 / sq ft**
18-inch	\$7 - \$9 / sq ft*	\$12 - \$15 / sq ft*	\$18 - \$22 / sq ft**	\$25 - \$35 / sq ft**
24-inch	\$9 - \$11 / sq ft*	\$16 - \$19 / sq ft*	\$22 - \$25 / sq ft**	\$35 - \$37 / sq ft**

*Addition of geotextile fabric to a marine mattress will increase the unit cost by \$0.05 - \$0.25 / sq ft (typically) depending on the fabric specification

**Installation rates and costs vary depending on the project size and anticipated degree of difficulty.

The pricing in each column is cumulative so the price in column 2 includes the cost for the previous column, plus the addition of materials and labor. The cost in columns 3 & 4 include all of the material and labor from the previous columns and the cost of the installation.

It is understood that site conditions in some locations require additional steps and/or equipment. No cost consideration is given to such scenarios in this table as each situation is different. The installation costs represented here are reasonable for typical installations.

Allowable Length of Marine Mattress for Safe Lifting ($fs \geq 4$)

Mat Thickness	MUX200	MUX300
12-inch (30cm)	35-ft	45-ft
18-inch (45cm)	24-ft	38-ft
24-inch (60cm)	17-ft	23-ft
	BX3030	BX4040
6-inch (15cm)	24-ft	28-ft

Note that it is not practical to make 6-inch thick mattresses out of UX Geogrids. Due to the aperture size and configuration of the Uniaxial (UX) geogrid, the stone gradation recommended for best performance should include a D50 that is approximately 2-inches and it is recommended that the stone be graded (i.e., 2" to 4") such as to facilitate tight packing and limit void space within the compartments (which is key to limiting interstitial velocity and enhancing performance of the mattress system).

Stone Recommendations

Unless otherwise directed by the engineer of record, Industrial Fabrics recommends filling marine mattresses with durable, angular granite (density 2.6+). We realize that this is not always possible or economically feasible. In such cases alternate materials including lime rock and crushed concrete may be approved. Lighter, less durable materials may impact the performance of the mattresses if they are to be used in wave conditions or high velocity applications.

Below are the recommended gradations for 6-inch and 12-inch mattresses.

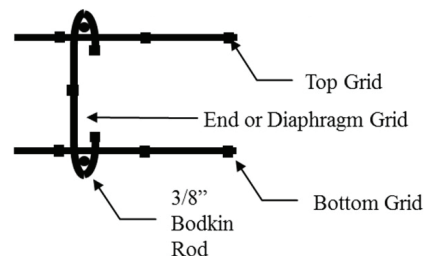
12-INCH MARINE MATTRESS STONE RECOMMENDATION:

The D50 of stone used shall be approximately 2-inches to 2.5-inches. The minimum diameter shall be 1-inch across the smallest dimension of the stone. The maximum diameter of stone used shall be 4 inches. Aggregate meeting the AASHTO #1 specification is generally acceptable as fill for 12-inch marine mattresses.

6-INCH MARINE MATTRESS STONE RECOMMENDATION: HORIZONTAL FILLING GEOTEXTILE LINED

<u>Diameter (inches)</u>	<u>%, by Weight, Passing</u>
3	100
2-1/2	90-100
2	35-70
1-1/2	0-15
3/4	0-5

Expanded Section at Diaphragm



Filling

Mattresses fabricated from UX Geogrids must be filled in the vertical position. This is a detail that cannot be changed due to the way that the baffles are connected to the top and bottom panels via bodkin rod connectors prior to filling (see diagram below). BX Mattresses can be fabricated for vertical or horizontal filling.

It is common for contractors to fabricate their own frames for marine mattress filling, but Industrial Fabrics does offer frames for lease. Detailed instructions can be found in the marine mattress filling and installation guide at www.baselok.com/marine

Geotextile

The use of geotextile in conjunction with a marine mattress is recommended in most cases. It is important to note that the engineer is responsible for selecting the appropriate geotextile. Commonly used geotextiles include non-woven (8, 10 and 12 ounce per sq yd) and, in some cases woven monofilament.